## Continuous Random Variables

A continuous random variable can take on any numerical value within a given range.
Let $X$ be a continuous random variable.
We associate a probability density function (or a probability model) $p(x)$ with

$$\int_{-\infty}^{\infty} p(x)\,dx = 1.$$

The probability that $X$ takes on a value within an interval can be viewed as an area under the curve of $p(x)$.
That is, for some interval $a \le x \le b$,

$$P(a \le X \le b) = \int_a^b p(x)\,dx.$$

The cumulative distribution function is

$$F(t) = P(X \le t) = \int_{-\infty}^{t} p(x)\,dx.$$

We can compute expected value and variance:

$$\mathbb{E}[X] = \int_{\mathbb{R}} x \cdot p(x)\,dx, \quad \mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Ex. The number of days before a computer breaks down is a continuous random var. with PDF

$$p(x) = \begin{cases} \lambda e^{-x/100} & x \ge 0, \\ 0 & x < 0. \end{cases}$$

- Compute $\lambda$.
- Find the probability that it will last between 50 and 150 days.

- Since $p(x)$ is a PDF: $\int_{\mathbb{R}} p(x)\,dx = 1$

$$\int_0^{\infty} \lambda e^{-x/100}\,dx = \lambda\left[-\frac{1}{1/100} e^{-x/100}\right]_0^{\infty}$$

$$= -100\lambda[0 - 1] = 100\lambda = 1.$$

Thus, $\lambda = 1/100$.

- This is just:

$$P(50 < X < 150) = \int_{50}^{150} \frac{1}{100} e^{-x/100}\,dx$$

$$\approx 0.3834005\ldots.$$

## Uniform Distribution $U(a, b)$

The PDF is

$$p(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \le x \le b, \\ 0 & \text{otherwise.} \end{cases}$$

The CDF is

$$F(t) = \begin{cases} 0 & \text{if } t < a \\ \frac{t-a}{b-a} & a \le t \le b \\ 1 & \text{if } t > b \end{cases}$$

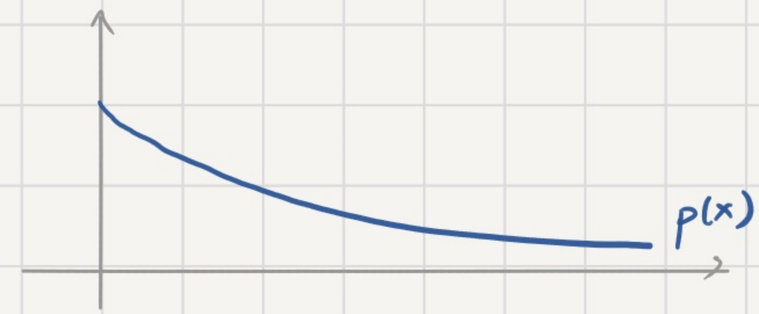
The expected value is

$$\mathbb{E}[X] = \frac{a+b}{2}.$$

The variance is

$$\mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

## Exponential Model $\mathrm{Exp}(\lambda)$




The PDF is

$$p(t) = \begin{cases} \lambda e^{-\lambda t} & \text{if } t > 0, \\ 0 & \text{otherwise.} \end{cases}$$

The CDF is given by

$$P(X \le t) = \int_0^t \lambda e^{-\lambda x}\, dx = -e^{-\lambda x}\Big|_0^t$$

$$\Rightarrow F(t) = \begin{cases} 1 - e^{-\lambda t} & \text{if } t > 0 \\ 0 & \text{otherwise.} \end{cases}$$

The expected value is $\mathbb{E}[X] = 1/\lambda$.

$$\mathbb{E}[X] = \frac{1}{\lambda}.$$

The variance is

$$\mathrm{Var}(X) = \frac{1}{\lambda^2}.$$